Kaa: An Easy-to-Use Platform for Building IoT Solutions

Kaa is a multi-purpose middleware platform for the Internet of Things that allows building complete end-to-end IoT solutions, connected applications, and smart products.

Kaa is a 100 per cent open source middleware platform for building end-to-end IoT solutions, connected applications and products. The Kaa IoT platform is licensed under Apache 2.0. Kaa takes care of all the backend heavy lifting and allows vendors to concentrate on maximising their product's unique value. It is horizontally scalable, fault-tolerant, and provides a broad set of features. Kaa was created for IoT companies and individuals interested in retaining ownership of the entire technological stack that they create.

An overview of Kaa

Kaa is a multi-purpose middleware platform for the IoT that allows you to create complete and smart applications. The Kaa platform provides an open, feature-rich toolkit for the IoT product development process and thus dramatically reduces the associated costs, risks and time-to market. It offers a set of out-of-the-box enterprise-grade IoT tools that can be easily plugged in and implemented in a large majority of the IoT use cases.

There are numerous architectural specifics that make IoT development with Kaa fast and easy. First, Kaa is hardware-agnostic and thus compatible with virtually any type of linked device, sensor and gateway. It also provides a clear construction of IoT features and extensions for different types of IoT applications. These can be used almost as plug-and-play modules with minimal addition of code on the developer's part. Kaa introduces standardised methods that enable integration and interoperation across connected products. And it is designed to be robust, flexible, and easy-to-use and deploy.

Kaa's features

Some features of Kaa that enable key IoT capabilities are listed below:

1. Analyses user behaviour and delivers targeted notifications.
2. Manages an unlimited number of connected devices.
3. Performs real-time device monitoring.
4. Performs remote device provisioning and configuration.
5. Creates cloud services for smart products.
6. Sets up cross-device interoperability.
7. Distributes over-the-air firmware updates.
8. Collects and analyses sensor data.
9. Performs A/B service testing.

How it works

Kaa enables data management for connected objects and the backend infrastructure by providing the server and endpoint SDK components. The SDKs get embedded into your connected device and implement real-time bi-directional data exchange with the server. Kaa SDKs are capable of being integrated with virtually any type of connected device or microchip.

The Kaa server provides all the backend functionality needed to operate even large scale and mission-critical IoT solutions. It handles all the